



Decoding Animal Internal States Using Wireless Implantable Accelerometers

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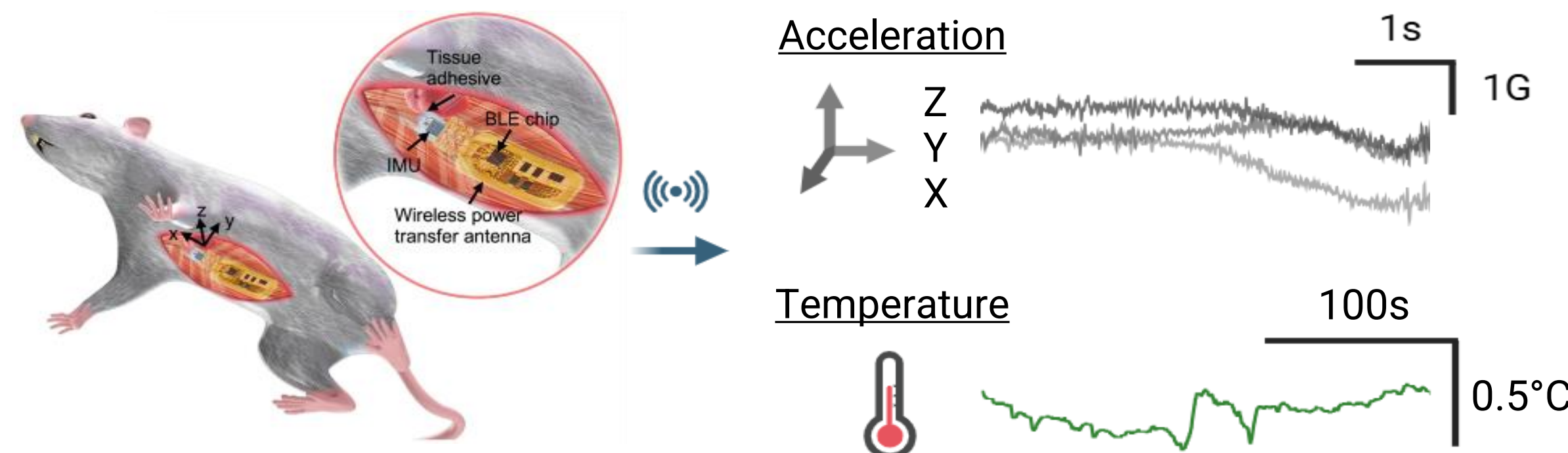
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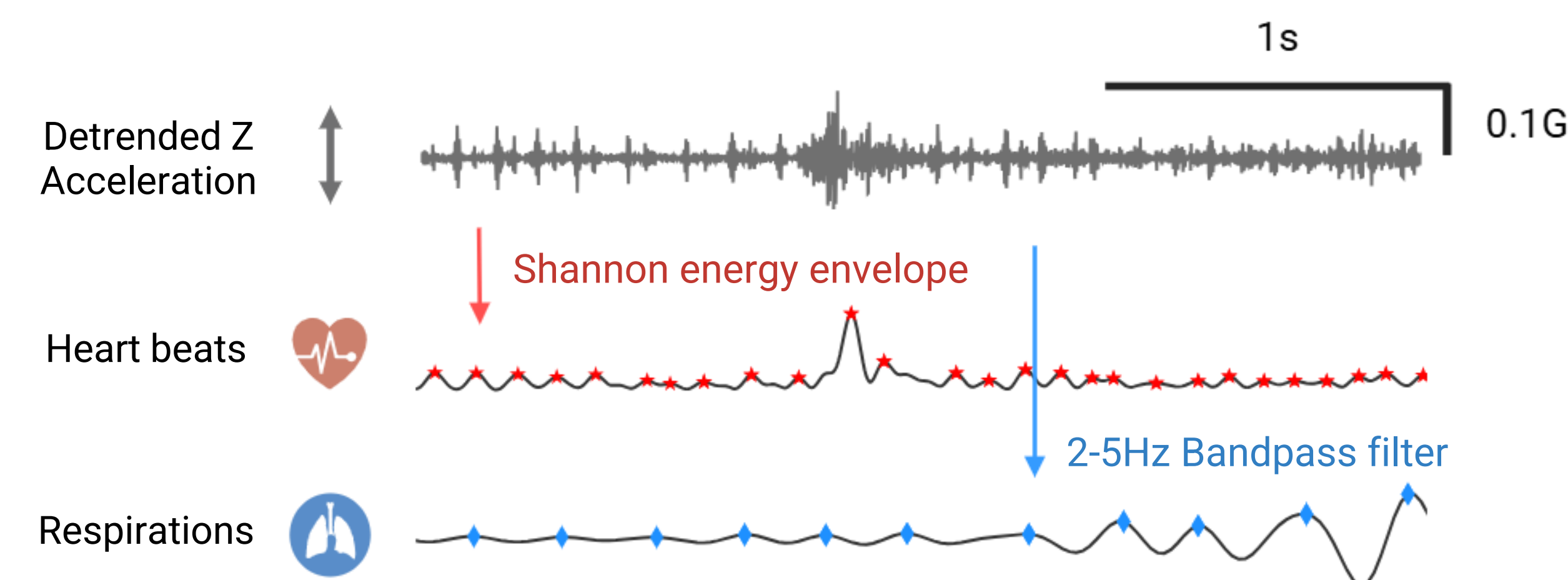
Background

- Rodent models are essential for studying neural mechanisms of behavior and disease
- Behavioral annotation is time-intensive, subjective, and often misses internal states (e.g., aggression, anxiety, drug-craving)
- Peripheral physiology (HR, RR, temperature) and behavioral motion provide objective markers of internal state
- Traditional recording methods constrain behavior; wireless methods enable naturalistic measurement
- We combine lightweight, implantable sensors and machine learning to decode behavioral and internal states

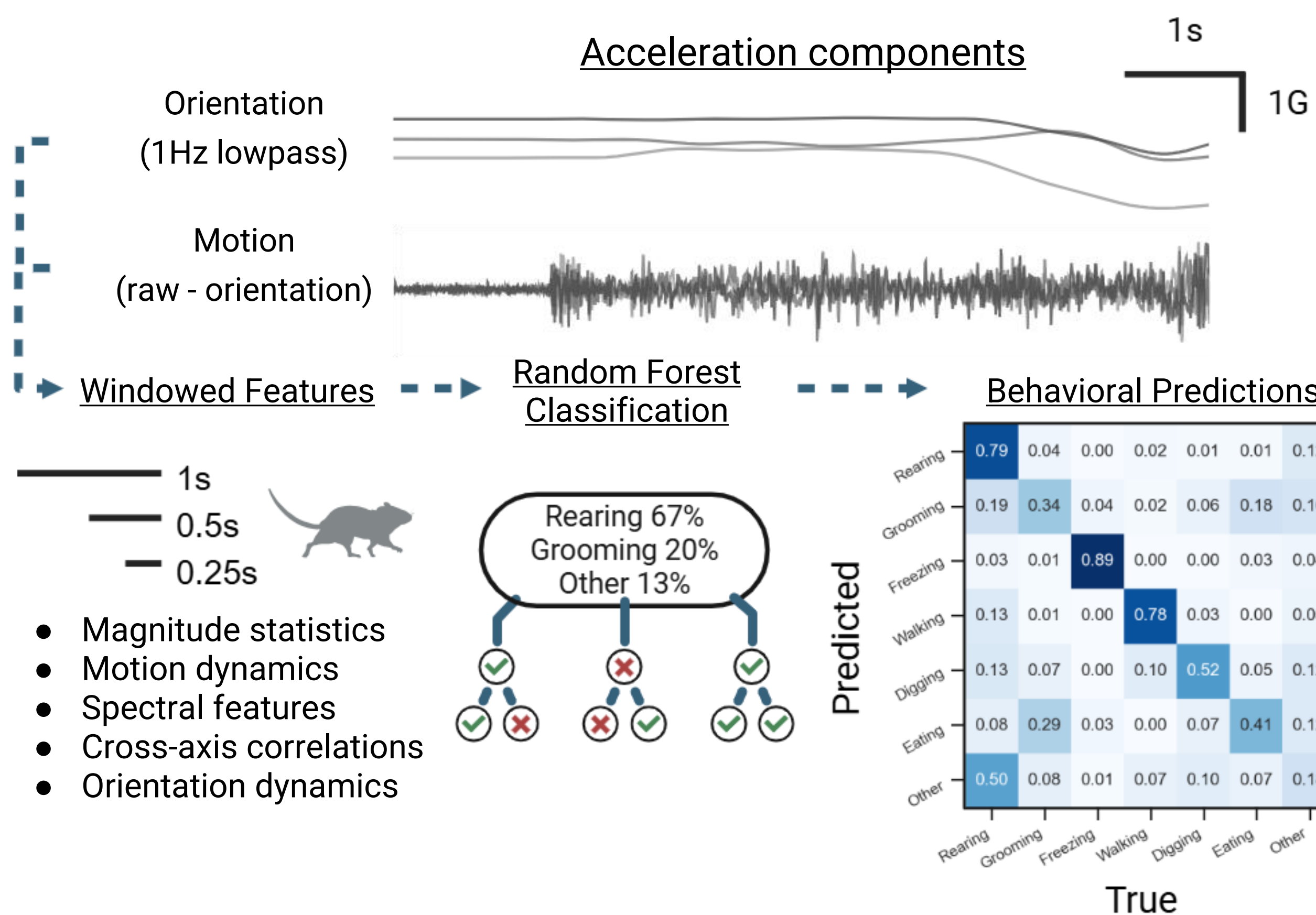
Wireless Implant Captures Motion and Temperature



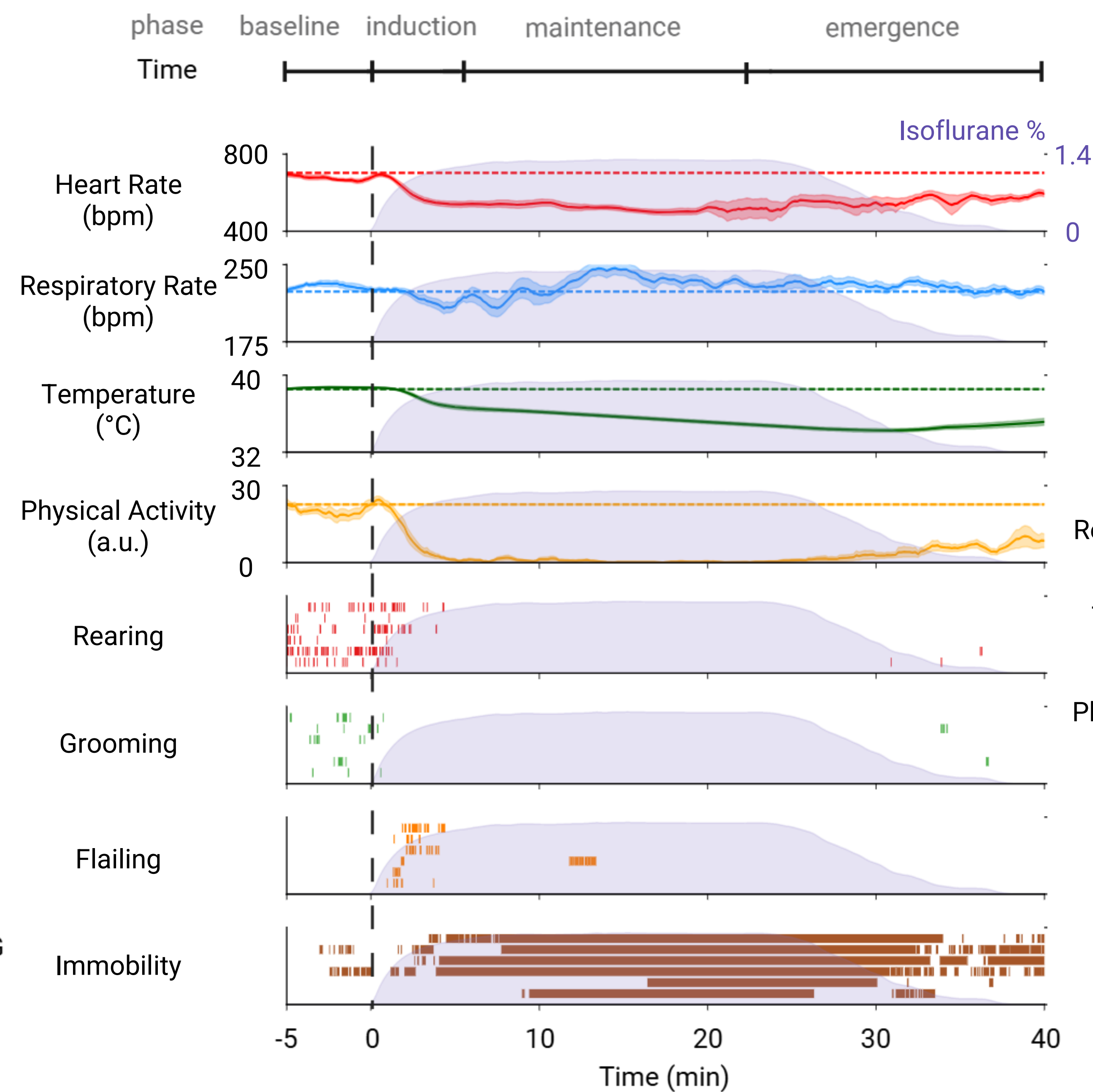
Physiological Analysis



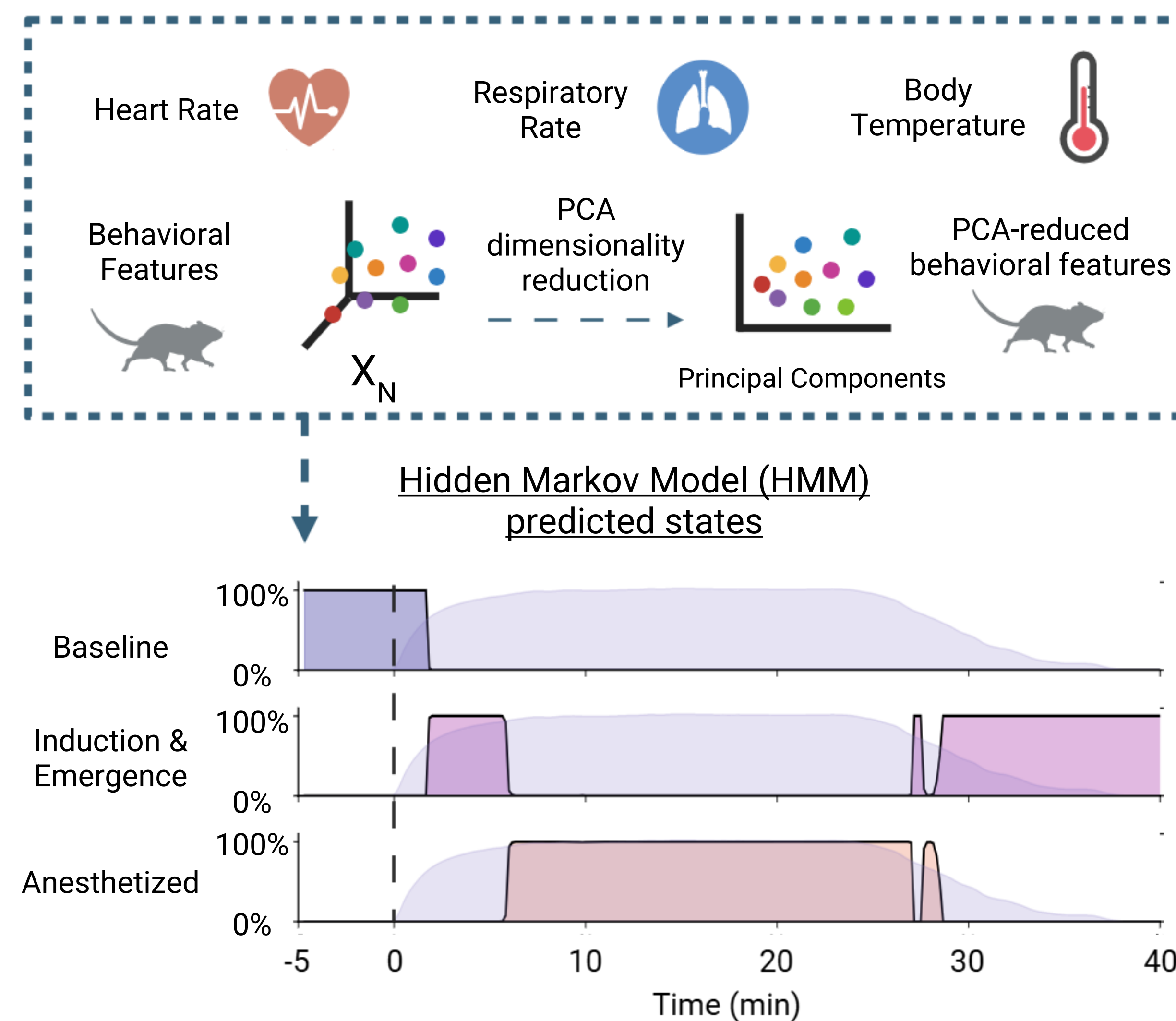
Supervised Behavioral Classification



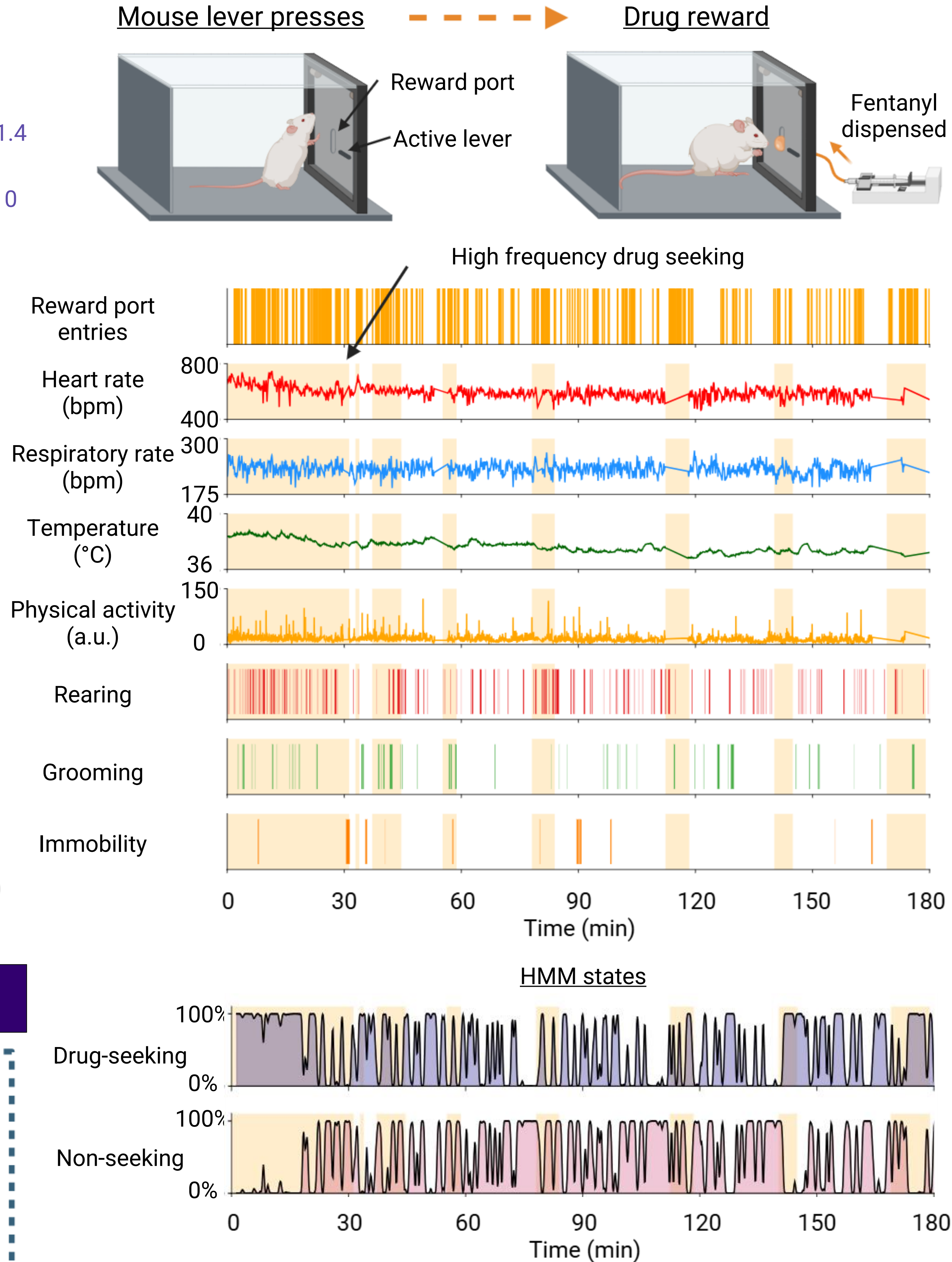
Internal State During Isoflurane Anesthesia



Hidden Markov Internal State Classification



Internal State During Fentanyl Self-Administration



Conclusions

- Wireless accelerometers capture both behavioral motion and physiological signals
- Machine learning enables classification of behavioral and internal states
- Unsupervised models reveal latent internal states underlying observable behavior
- Fentanyl induces systematic shifts in physio-behavioral state dynamics
- Real-time state detection enables closed-loop neural intervention using optogenetics

Acknowledgements

- Devices and physiological analysis pipeline from Ouyang, Wei et al., 2024
- Thank you to my mentor Sam Golden and all the members of the Golden Lab!
- Mary Gates Endowment